

 PROJECT OVERVIEW



THE CHALLENGE

Predict five Big Five trait scores from five short, open-ended situational-judgment answers — scored by mean Pearson r .



THE DATA

1,688 respondents · Train 1,088 / Dev 300 / Test 300

Targets 1–5 Likert; traits intercorrelate (N negative) → modeled per-trait.



PRIVATE-TEST MEAN r

0.3215

Train-CV 0.3635 · Public-Dev 0.360

E .41

A .39

C .28

O .27

N .25

BEATS 2019 1ST PLACE +0.061 (~23%)

 THE STACKED SOLUTION

Zero-shot LLM bases + classical bases, combined by a leakage-safe meta-learner



LLM 01

Haiku 4.5 scorer

×4 prompt framings



LLM 02

Sonnet 4.6 judge

independent 2nd read



LLM 03

Behavioral feats

16 structured signals



LLM 04

Role-play BFI-2

30-item questionnaire



CLASSIC 05

e5-large SVR

per-prompt + all-text



CLASSIC 06

TF-IDF + feats

Ridge + HistGBM



META-LEARNER

per-trait Ridge ($\alpha=4$) · Train 5-fold OOF stacking

WHAT SET IT APART

- 01 Frontier LLMs as zero-shot extractors
- 02 Multi-prompt + independent second judge
- 03 Role-play questionnaire simulation

 RESULTS & STANDINGS

2019 FINAL STANDINGS

private test · mean Pearson r (higher is better)

RANK	TEAM	MEAN r
1	This solution ★	0.3215
2	Walmart · Nat. Selection	0.2602
3	BGSU · Procrustination	0.2478
4	Michigan State	0.2325
5	USAA · PI-RATES	0.2293



THE MARGIN: +0.061 over the 2019 winner — about 2× the entire top-4 spread (0.031).



TRAINING DETAILS

Embeddings	e5-large-v2 · 1024-d · L2-norm
Kernel head	SVR · RBF · per-prompt + all-text
Sparse head	TF-IDF char+word → Ridge

WHAT DIDN'T WORK

- ▶ SBERT MiniLM — lost to TF-IDF (0.14 vs 0.22)
- ▶ bge / gte embedders — weaker than e5-large
- ▶ Cross-trait meta — CV 0.34 → Dev 0.28

KEY LEARNINGS



Zero-shot LLMs break the ceiling

Frontier priors beat anything trainable on 1,088 rows.



Leakage-safe by construction

Zero-shot bases can't leak; meta fit on Train OOF only.



Openness finally moved

The hardest trait went 0.06 → 0.27 vs the baseline.



Ensembling beats architecture

Diverse bases + stacking drove the gain, not one model.